Yanhe Ren

Machine Learning / Research Intern | Summer 2027 (May-Aug)

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EDUCATION

Arizona State University | Ph.D. in Statistics — Aug 2025 - Present | Expected May 2029

University of California, Irvine | M.S. in Electrical and Computer Engineering — Sep 2022 - Jun 2024

East China University of Science and Technology | B.Eng. in Automation — Sep 2018 - Jun 2022

RESEARCH EXPERIENCE

FlowPINN: One-Step Generative Modeling | *Under review*

Arizona State University | Doctoral Research | Advisor: Shiwei Lan — Aug 2025 - Present

- Enabled single-evaluation generation with FlowPINN by learning a physics-informed neural flow map through minimization of the flow ODE residual at the map's output.
- Achieved **117x faster** MNIST sampling with comparable FID (24.29 vs. 24.23) by replacing 100-NFE CFM solves with single-evaluation FlowPINN-L sampling on a UNet backbone.
- Achieved one-step CIFAR-10 **FID of 8.69**, compared with 9.88 for MeanFlow and 23.65 for Shortcut.

Uncertainty Quantification for Flow Matching | *Under review*

Arizona State University | Doctoral Research | Advisor: Shiwei Lan — Aug 2025 - Present

- Estimated per-sample numerical error relative to each model's ODE solution using the sampler's Jacobian and two scales calibrated offline per sampler and step count.
- Developed a probabilistic ODE solver for offline calibration and parallel trajectory refinement using a Gauss-Markov prior and Gauss-Newton conditioning across time nodes in log-SNR time.
- Reduced CIFAR-10 FID from **11.05 to 5.13** and halved MNIST median ODE-endpoint error from **7.2 to 3.6** by refining FlowPINN samples in three probabilistic-solver rounds.

SMILES: Mobile Data Collection and Analytics

University of California, Irvine | Research & Data Analyst | Annie Qu's Lab — Apr 2024 - Apr 2025

- Supported the collection of **31K+ EMA responses** for longitudinal stress, sleep, and activity research by building and releasing a Flutter-based iOS/Android app for the SMILES study.
- Developed a Flask/MySQL data-ingestion and analytics backend, deployed with Gunicorn and Nginx, connecting mobile data collection with research reporting.
- Integrated Oura Ring data, implemented SQL constraints and outlier checks, and delivered de-identified datasets and cohort dashboards used in weekly research reports.

Other projects at UC Irvine

- Prototyped a multi-turn medical RAG assistant using LLaMA, FAISS, and M2-BERT embeddings; assessed retrieval with Recall@5/MRR and audited unsupported responses through an offline evaluation harness.
- Curated **1.7M arXiv abstracts** with titles, categories, and timestamps to support dynamic topic modeling.

SELECTED PUBLICATION

Context-Sensitive Stress Prediction from Wearable and EMA Data for Just-in-Time Adaptive Interventions.
Cadence Pinkerton, **Yanhe Ren**, Yuqing Guo, Annie Qu
STAI-X 2026, Poster Track.

TECHNICAL SKILLS

Programming: Python, SQL, R

Methods: Generative modeling, uncertainty quantification, Bayesian statistics, retrieval-augmented generation

Tools: PyTorch, Linux, Slurm, Git, MySQL, Flask